

✈️ Flight Filters

Flight Phase

Climb ✕

Cruise ✕

✕ ▼

Descent ✕

Ground ✕

Hybrid Risk Category

Critical ✕

High ✕

✕ ▼

Low ✕

Medium ✕

OCC Alert Level

Critical ✕

High ✕

✕ ▼

Medium ✕

Routine ✕

Origin Country

Afghanistan ✕

Argentina ✕

Austria ✕

Bahamas ✕

Bangladesh ✕

Intelligent Flight Operations Analytics Dashboard

Hybrid AI-Based Risk Assessment and Information Overload Mitigation

This dashboard supports operational decision-making for an Operations Control Centre (OCC).

The system integrates:

- Gaussian Mixture Model (Behaviour Analysis)
- ELM Autoencoder (Anomaly Detection)
- Hybrid AI Risk Assessment
- Smart Alert Prioritisation
- Explainable AI (SHAP)

to identify high-risk flights, prioritise operational alerts, and reduce information overload.

✓ Datasets loaded successfully!

Executive Overview

 Total Flights	 Average Risk Score	 High Risk Flights	 Critical Alerts
7,516	35.03	1,364	53

Flight Risk Analysis

Hybrid Flight Risk Score Distribution

200

Hybrid Risk Category Distribution

